

Performance Report: Deepfake Audio Detection

Project: Deepfake Audio Detection

Objective: Classify speech recordings as either **Genuine (Human)** or **Deepfake (AI-Generated)**.

Model Overview

The model uses a spectrogram-based CNN approach for audio classification. Raw audio waveforms are first converted into fixed-length 2-second signals at 16 kHz. These waveforms are then transformed into STFT-based spectrograms, resized to 128 × 128, standardized, and passed to a CNN model for classification.

Evaluation Method

The trained model was evaluated on the unseen test dataset. The final prediction was made using the selected Equal Error Rate threshold instead of the default 0.5 threshold. This threshold helps balance false acceptance and false rejection errors.

Final Test Metrics

| Metric                 | Value  |
|------------------------|--------|
| Accuracy               | 92.83% |
| Equal Error Rate (EER) | 7.17%  |
| Macro F1-Score         | 92.83% |
| Weighted F1-Score      | 92.83% |
| Fake Class Accuracy    | 92.83% |
| Real Class Accuracy    | 92.83% |

Result Summary

The model achieved strong performance on the test dataset, with an overall accuracy of **92.83%** and an Equal Error Rate of **7.17%**. Both fake and real classes achieved **92.83% per-class accuracy**, showing that the model performed consistently across both categories.

The low EER value indicates that the model maintained a good balance between false positives and false negatives. Overall, the results show that the spectrogram-based CNN model is effective for distinguishing between genuine human speech and AI-generated deepfake audio.